

Mansi Sinha

📍 Delhi, IN ✉️ mansi15aug@gmail.com 📞 +91 93542 03506 📁 Portfolio in LinkedIn 🐙 GitHub

Technologies

Languages/Libraries: Python, SQL, Pandas, NumPy, Matplotlib, Seaborn

ML/AI: Scikit-learn, TensorFlow, Keras, LangChain, CrewAI

LLMs/Gen AI: Gemini 2.5, OpenAI API, RAG pipelines, Prompt Engineering

MLOps/Cloud: MLflow, Docker, Git/GitHub, FastAPI, CI/CD, Jupyter, AWS

Techniques: Deep Learning, Autoencoders, Recommendation Systems, Multi-Agent Systems

Projects

SHE – AI Women’s Health Platform (Founder & Developer) (Ongoing)

[Live Demo](#) 

- Building an AI-powered women’s health platform focused on blood report analysis & personalized health insights.
- Designing OCR and NLP pipelines to extract biomarkers from laboratory reports (PDFs & images) and convert them into structured health data.
- Developing backend services using FastAPI for report processing, health reasoning workflows, and AI-powered conversational assistance.
- Architecting secure report storage, historical health tracking, and personalized insight generation. Architecting secure report storage, historical health tracking, and personalized insight generation.
- Creating an end-to-end system integrating OCR, LLMs, databases, and health-focused AI workflows for user-friendly report interpretation.
- Technologies: Python, FastAPI, OCR, NLP, PostgreSQL, OpenAI API, Git, Next.js

Real-Time Fraud Detection

[link](#) 

- Imbalanced classification (3.5% fraud rate) on IEEE-CIS dataset (590k transactions, 434 features) using XGBoost + Isolation Forest ensemble — achieved [PR-AUC score] with [X]% fraud recall.
- Engineered [X] features from raw transaction data including card velocity, time-based signals, and amount z-score — SHAP TreeExplainer revealed [top feature] as strongest fraud predictor.
- Served predictions via FastAPI REST endpoint achieving with Kafka stream simulation.

Financial Document Analyzer

[link](#) 

- Developed a production-ready multi-agent AI system using CrewAI and FastAPI to analyze financial documents and generate structured insights including risk assessment and investment recommendations.
- Designed a sequential agent pipeline (Verifier, Analyst, Risk Assessor, Advisor) ensuring contextual information flow and logical decision-making across tasks.
- Resolved complex dependency conflicts and refactored the system architecture, stabilizing the environment by managing packages like LangChain, OpenAI, and ChromaDB.

Experience

Freelance Data Science Projects

Remote

Independent / Remote

Apr 2024 – Dec 2025

- Worked on independent machine learning and data analysis projects while pursuing MSc in AI & ML.
- Built predictive models using Python and Scikit-learn for classification and regression tasks.
- Performed data cleaning, feature engineering, and exploratory data analysis on real-world datasets.

Data Analytics Intern

Remote

Alfido Tech

Oct 2023 – Jan 2024

- **Data analysis** tasks were performed using **SQL** to extract, clean, and manipulate datasets.
- Generated comprehensive reports and **Tableau dashboards**, improved data visualization for stakeholders.
- Worked extensively with **SQL joins to integrate and analyze data from multiple sources**, ensuring data consistency and accuracy.

Data Science Intern

Solar Secure Solutions

Remote

Jul 2023 – Sept 2023

- Complex data sets analyzed and interpreted, providing actionable insights that were improved by 20%.
- Developed **statistical models and machine learning algorithms** (e.g. regression, classification, decision trees) to extract meaningful patterns, enhancing the predictive accuracy by 15%.

Education

Bharati Vidyapeeth's College of Engineering (GSSIPU, Delhi)

Jul 2023

B.Tech. in Electronics & Communication CGPA: 9.17/10 [🔗](#)

Scaler Neovarsity (Woof University, ECTS Accredited (EQF7))

Dec 2025

M.Sc. in Computer Science (AI & ML) GPA: 3.9/4.0 [🔗](#)